

The Energy Balance Project

Homeostatic Field Model and Local Actor-Motion Law

Foundation Document — Version 2.5 (EBU Accounting: Naive vs Guarded under Adversaries)

Status: an EBU accounting/incentive layer added on top of the frozen V2.4 threshold-aware physics (baseline tag v2.4.0). The physics equations are unchanged. Backed by 9 new ledger-property tests (33 tests total) run before any experiment.

Headline (falsifiable) result. A naive EBU ledger is exploitable in specific, enumerated ways: adversaries maximising their naive balance destroy the system entirely (0% viable, 32/32 sources dead) while being paid a huge amount of credit (~730,000). A guarded ledger — live-state causal credit against the threshold-aware burden, issued sequentially, with symmetric burden-increase debits and explicit debits for transport loss and irreversible extraction — closes or bounds every tested attack and keeps adversaries at 100% viability: maximising guarded EBU coincides with doing real homeostatic work. Residual, non-exploitable credit is characterised.

1. Architecture

The EBU layer may observe actions and their verified consequences, maintain per-actor accounts, and influence selection among physically admissible actions. It must NOT change regeneration, transport, demand, or conservation. The physics is exactly V2.4's threshold_penalty rule and its threshold-aware burden B_R. A property test confirms the physical trajectory is identical under exact floating-point equality (no rounding) whether the ledger is off or guarded — the layer is observational unless it drives selection.

2. The Guarded Ledger

Credit is the live-state reduction in the threshold-aware burden B_R around each action, issued SEQUENTIALLY against the running state so natural inflow/regeneration are never credited and two actors cannot be paid for the same reduction (telescoping). Burden INCREASES are debited symmetrically; transport dissipation and irreversible extraction below the Allee threshold are debited explicitly:

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credit_a = max(0, B_R(before_a) - B_R(after_a))          (issued credit, >= 0)
dEBU_a   = B_R(before_a) - B_R(after_a) - lambda_L L_a - lambda_F F_a
          = credit_a - max(0, -(B_R diff)) - lambda_L L_a - lambda_F F_a
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L_a = (1-eta) q + c0          transport dissipation of the action
F_a = increase in (A_i - x_i)_+ at a regenerative source i  (extraction below Allee)
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The symmetric burden-increase debit is a deliberate refinement of the plain $C_a = \max(0, .)$ formula: without it, an actor could do good, undo it, and keep the credit. With it, the signed burden terms telescope exactly, giving the account identity

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sum_a dEBU_a = B_R(x_before) - B_R(x_after) - (explicit cost debits)
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where the explicit cost debits are only the transport-dissipation and irreversible-extraction terms ($\lambda_L L_a + \lambda_F F_a$). Burden increases are already captured inside the telescoping B_R term, so they must NOT be subtracted a second time. This identity makes round-trips and damage-then-repair non-positive. The naive ledger, by contrast, credits apparent improvement in the PLAIN burden against the fixed pre-tick field, with no debits — so it credits natural regeneration, double-counts across actors, and is blind to transport loss and regenerative-reserve sacrifice.

3. Tests Before Experiments

All nine guarded-ledger properties hold (part of a 33-test suite):

Property	Result
No action → zero credit	PASS
Natural regeneration not credited: guarded credit = action-only reduction (1.75); naive inflates to 13.42	PASS
Telescoping: issued credit == actual B_R reduction (no double credit)	PASS
Perfect (lossless) round trip returning to start → net EBU = 0	PASS
Pointless lossy move (in-band) → strictly negative EBU	PASS
Damage then exact repair → net EBU ≤ 0	PASS
Splitting one transfer into many → no increase in reward	PASS
Sum of account changes == issued credit - issued debit	PASS
Physics trajectory identical (exact float equality) when the ledger is observational	PASS

4. Four Comparisons

Closed Allee economy (8x8, only supply is regeneration), 800 ticks, source shock at tick 400. The first three share one physics trajectory (EBU observational); the last two let actors choose actions to maximise their own EBU balance.

Selection / ledger	viable %	dead	B_R mean	issued credit	issued debit	net EBU
physics / none	100.0	0/32	93.6	0	0	0
physics / naive (obs)	100.0	0/32	93.6	1,682	0	1,682
physics / guarded (obs)	100.0	0/32	93.6	5,994	71	5,922
adversarial / naive	0.0	32/32	2,558	730,913	0	730,913
adversarial / guarded	100.0	0/32	96.3	5,513	80	5,433

The naive adversary earns ~730,000 EBU while collapsing every source — a total decoupling of reward from health. The guarded adversary cannot do this: maximising the guarded balance drives genuine homeostatic work, so it keeps 100% viability and earns only bounded, earned credit (comparable to the observational guarded case). The guarded adversary's positive net EBU (5,433) is real work, not gaming.

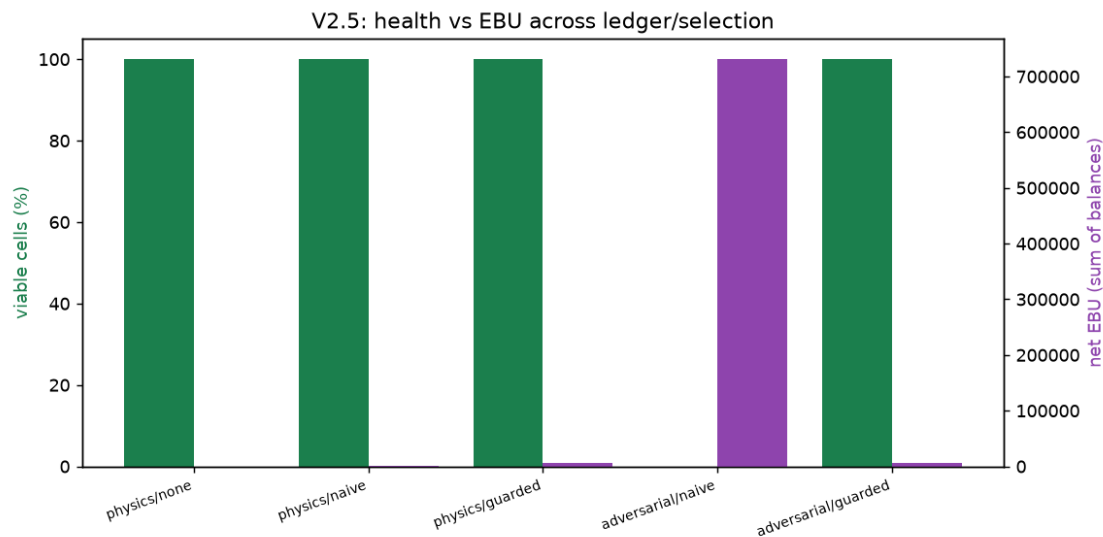


Figure 1. Health (green, left axis) vs net EBU issued (purple, right axis). Only adversarial/naive decouples them: zero viability with a ~730k EBU spike. Guarded keeps adversaries healthy; its EBU bars are negligible on this scale.

5. Attack Suite

Net EBU an actor earns by executing each known gaming strategy, under each ledger. Guarded makes every attack non-positive or strictly smaller than naive.

Attack	naive EBU	guarded EBU	outcome
Back-and-forth (round trip)	4.94	0.89	reduced; residual is real net rebalancing (a true return-to-start nets 0)
Damage then repair	0.00	0.00	no gain either way (lossless)
Split one transfer into four	+11.36	-0.06	closed: splitting pays extra transport, no gain
Sacrifice regen (harvest below A)	0.00	-38.02	closed: naive is free, guarded strongly debits
Claim natural regeneration	13.42	1.75	closed: guarded credits only the action's live effect

Duplicate credit (two actors, one fix)	16.00	8.00	closed: guarded telescopes to the real reduction
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Honest characterisation of residuals: the back-and-forth guarded value (0.89) is positive only because that specific lossy sequence does not return the system to its start — it performs a small net rebalancing, which is genuine work; a true return-to-start round trip nets exactly zero (property test). The sacrifice-regen attack shows the sharpest contrast: naive assigns it zero cost (plain burden is blind to a source dropping below its Allee threshold while still above L), so under naive it is a free way to destroy sources; guarded debits it heavily.

6. Verdict

The meaningful result is not ‘EBU works’. It is that **naive EBU is exploitable in the enumerated ways, while guarded EBU closes those attacks without changing the physical homeostasis** (the physics trajectory is provably identical when the ledger is observational, and the guarded adversary stays at 100% viability). Maximising the guarded balance is aligned with system health by construction: credit is the verified live-state burden reduction, and every way of manufacturing apparent value — churning, splitting, claiming regeneration, double-claiming, or sacrificing regenerative stock — is either uncredited or debited.

What is NOT claimed. The attack set is finite; passing it is not a proof that no exploit exists. Guarded credit is local (per action, on the two affected cells); collusion, multi-hop laundering, and slow baseline manipulation across many ticks were not adversarially searched. The guarded adversary still earns positive EBU for real work — which is intended, not a leak.

7. Limitations and Next Steps

1. Search for exploits automatically (a learning adversary) rather than testing a hand-written list, and add collusion / multi-actor laundering scenarios.
2. Characterise the guarded residual formally: bound net EBU by realised, persistent B_R reduction minus dissipation.
3. Stress the credit-locality assumption with long-range transport and shared sources.
4. Only if guarded survives should EBU be allowed to influence physics-relevant parameters; for now it stays an incentive/selection layer over the fixed V2.4 baseline.

Version status. V2.5 adds an EBU layer over the immutable v2.4.0 physics. The physics is unchanged and re-verified. V2.4 result captures are archived under results/v2.4/ with a manifest. The next revision should add an automated adversary and collusion tests before trusting the guarded ledger in any deployment sense.